1. Scope & Planning

STEP 1 - Scope & Planning | CNC Machining - Ti-6Al-4V Femoral Stem

Field	Detail
Process Name	CNC Machining - Ti-6Al-4V Femoral Stem
Scope Start	Raw Ti-6Al-4V Bar Stock Receipt
Scope End	Final Dimensional Inspection / Release
Device Classification	Class III - PMA (21 CFR 814)
Standard Applied	AIAG-VDA FMEA 4th Edition 2019
Regulatory Framework	21 CFR 820 (QSR) | ISO 13485:2016
Device Description	Cementless Ti-6Al-4V Femoral Stem - Implantable Hip Prosthesis
Team Members	Quality Engineer, Process Engineer, Supplier Quality, R&D;
FMEA Facilitator	[Your Name] - Quality Engineer
FMEA Number	PFMEA-HIP-001
Revision	Rev A
Date	2026-06-22
Next Review Date	6 months post-transfer validation
Status	Draft - Internal Review

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485

2. Structure Analysis

STEP 2 - Structure Analysis | System Hierarchy Breakdown

System Level	Element Name	Function Contribution
Level 1 - System	Manufacturing Cell	CNC Machining production environment
Level 2 - Subsystem	CNC Turning Station	Primary material removal and geometry creation
Level 3 - Component	Cutting Tool (Insert)	Executes material removal; controls Ra and geometry
Level 3 - Component	Workholding Fixture	Maintains part datum and prevents movement
Level 3 - Component	Coolant Delivery System	Controls chip evacuation and thermal stability
Level 3 - Component	CNC Part Program	Defines toolpath, feeds, speeds, and tolerances
Level 3 - Component	Operator / Setup Tech	Executes setup, monitors process, responds to alarms
Level 4 - Interface	CMM Inspection Station	In-process and final dimensional verification
Level 4 - Interface	Electronic Traveler (DHR)	Enforces operation sequence and records results

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485

3. Function Analysis

STEP 3 - Function Analysis | Intended Functions & Requirements

Process Element	Intended Function	Requirement Satisfied
Cutting Tool (Insert)	Remove material to achieve OD +/-0.010 mm and Ra <= 0.8 um on taper surface	Device drawing tolerance stack; osseointegration surface finish spec
Workholding Fixture	Maintain part concentricity <0.005 mm TIR during all turning operations	Geometric dimensioning: perpendicularity and runout callouts
Coolant Delivery System	Deliver coolant at >=50 PSI to maintain <60C tool-chip interface temperature	Ti-6Al-4V machining process spec; tool life and metallurgical integrity
CNC Part Program	Execute validated toolpath per ECO-approved revision; enforce feed/speed limits	Validated manufacturing process (IQ/OQ/PQ); 21 CFR 820.70(i) software control
Operator / Setup Tech	Load correct program revision, set datum offsets, perform first-article inspection	Work instruction WI-CNC-001; operator qualification records OQR-2024
CMM Inspection Station	Verify OD, ID, taper angle, and length within print tolerance post-machining	Inspection plan IP-HIP-003; acceptance criteria per device drawing Rev G
Electronic Traveler (DHR)	Enforce operation sequence gate checks; capture lot/SN, operator, and tool data	21 CFR 820.184 Device History Record; ISO 13485 7.5.3 traceability

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485

4. Failure Analysis

STEP 4 - Failure Analysis | Failure Modes, Effects & Causes

Process Step	Failure Mode	Effect on Patient / Downstream Process	Severity (1-10)	Root Cause	Current Prevention Control	Current Detection Control
CNC Turning - OD	OD out of tolerance (oversize)	Implant-stem mismatch; potential intraoperative fracture or non-union - patient harm	8	Tool wear beyond limit; incorrect offset entry; thermal expansion of Ti	Insert change interval SOP; offset verification at setup	Post-op CMM inspection; SPC chart OD trend
CNC Turning - Surface	Surface roughness Ra > 0.8 um on taper	Reduced osseointegration; long-term implant loosening - re-operation risk	9	Worn or chipped insert; inadequate coolant flow; excessive feed rate	Profilometer check per lot; insert change per interval	Profilometer measurement per part; visual inspection
Workholding - Fixture	Fixture slip / part shift mid-cycle	Geometric error cascade (runout, perpendicularity OOS); scrap or undetected escape	7	Improper clamping torque; fixture wear; incorrect locating pin seating	Torque spec on setup sheet; periodic fixture calibration	First-article runout check; operator pre-run confirmation
Coolant Delivery	Coolant pressure drop / flow loss	Thermal damage to Ti substrate; metallurgical alteration; implant scrapped or escaped	9	Clogged filter; pump failure; hose leak; operator valve error	Coolant PM schedule; filter change interval	Visual observation only - no automated interlock (GAP)
CNC Program - Revision	Wrong program revision loaded (obsolete program)	Incorrect geometry machined; toolpath collision risk; patient dimension non-conformance	8	Paper traveler allows manual program selection; no enforced program lock	Program naming convention; operator training	Setup verification checklist; first-article measurement
Inspection - CMM Step	CMM inspection step skipped / bypassed	Non-conforming part released to sterilization and distribution - direct patient risk	7	Production schedule pressure; traveler signature forgery; inadequate gate control	Work instruction WI-CMM-002; supervisor sign-off	DHR review at final release - detection is end-of-line only (GAP)

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485

5. Risk Rating

AIAG-VDA 2019 Action Priority (AP) -- NOT Traditional RPN

AIAG-VDA 2019 replaces Risk Priority Number (RPN = SxOxD) with Action Priority (AP: High / Medium / Low). Rationale for medical devices: a failure with Severity 9 that is NEVER detected (D=10) must be actioned immediately regardless of occurrence rate - RPN arithmetic can mask this by averaging scores. AP weights Detection gaps more heavily, reflecting that an undetected failure reaching a patient is catastrophic. The AP lookup table (not multiplication) drives priority.

AP Lookup: S9-10+D7-10->H | S9-10+D4-6+O>=4->H | S7-8+D7-10+O>=4->H | All others->M or L

STEP 5 - Risk Rating | AIAG-VDA Action Priority Table

Failure Mode	S (Severity 1-10)	O (Occurrence 1-10)	D (Detection 1-10)	Action Priority	AP Rationale	Legacy RPN (Reference Only)
OD out of tolerance (oversize)	8	4	3	L	S=7-8 + D<=3: detection adequate for severity level	96
Surface roughness Ra > 0.8 um	9	3	2	M	S>=9 + D<=3: strong detection mitigates critical severity	54
Fixture slip / part shift mid-cycle	7	3	4	L	S=7-8 + D=4-6 + O<4: controlled risk	84
Coolant pressure drop / flow loss	9	3	5	M	S>=9 + D=4-6 + O<4: rare critical but detection acceptable	135
Wrong program revision loaded	8	2	3	L	S=7-8 + D<=3: detection adequate for severity level	48
CMM inspection step skipped	7	4	7	H	S=7-8 + D>=7 + O>=4: high-severity, frequent, poorly detected	196

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485

6. Optimization

STEP 6 - Optimization / Corrective Actions | Risk Reduction Plan

Initial AP	Failure Mode	Action Description	Action Owner	Target Date	Revised S	Revised O	Revised D	Revised AP
H	Coolant pressure drop / flow loss	Install in-line coolant flow sensor with machine interlock: auto-feed hold + alarm if PSI < 45. Validate per IQ/OQ protocol IQ-COOL-001. Update PFMEA and Control Plan CP-HIP-001.	Process Engineer	2026-Q3	9	2	2	M
H	Wrong program revision loaded	Implement electronic traveler (eDHR) with barcode-locked program selection. Only approved ECO revision loads; prior revisions auto-archived. Validate per 21 CFR 820.70(i). Training completion rate 100% before go-live.	Quality Engineer + IT	2026-Q4	8	2	1	L
M	Surface roughness Ra > 0.8 um	Reduce insert change interval from 150 to 100 parts; implement profilometer check every 50 parts (sample basis). Update PM schedule and WI-CNC-001 Rev B.	Quality Engineer	2026-Q3	9	2	1	M
H	CMM inspection step skipped	Add mandatory CMM gate-check in eDHR workflow: part cannot advance until CMM pass record is uploaded. CMM data auto-linked to DHR lot record. Supervisor override requires Level III QE approval.	Quality Engineer	2026-Q4	7	3	2	L

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485

7. Results Summary

STEP 7 - Results & Documentation Summary

pFMEA Metrics Summary (auto-computed)

Total Failure Modes Analyzed	6
Initial High (H) Action Priority	1
High AP After Actions Implemented	0
Medium (M) AP - Initial	2
Medium (M) AP - After Actions	2
Low (L) AP - After Actions	4
Total Corrective Actions Assigned	4
Actions with Owner Assigned	4
Actions with Target Date	4

Linked Quality System Documents

Document Type	Document Number	Title / Description	Status
Control Plan	CP-HIP-001 Rev A	CNC Femoral Stem - Machining Control Plan	In Review
Work Instruction	WI-CNC-001 Rev B	CNC Turning Setup & Operation - Ti-6Al-4V Stem	Released
Inspection Plan	IP-HIP-003 Rev A	CMM Dimensional Inspection - Femoral Stem	Released
IQ/OQ Protocol	IQ-COOL-001	Coolant Interlock System - Installation Qualification	Pending
OQ/PQ Protocol	PQ-CNC-STEM-001	Process Validation - CNC Machining Transfer	Planned
Risk Management	RM-HIP-2024-01	Device Risk Management File (ISO 14971)	In Review
DFMEA	DFMEA-HIP-001	Design FMEA - Cementless Femoral Stem	Released
SOP	SOP-QE-022	pFMEA Creation and Maintenance Procedure	Released

Archive Location: [PLM System] -> Projects -> HIP-STEM-TRANSFER-2024 -> FMEA -> PFMEA-HIP-001_RevA.xlsx

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485

8. ISO 14971 Bridge

ISO 14971 RISK BRIDGE - best-effort mapping, QE review REQUIRED

Failure Mode (pFMEA)	Hazard	Hazardous Situation	Harm	Severity (S)	P1 (Prob. of hazardous situation)	P2 (Prob. of harm)
OD out of tolerance (oversize)	[derive]	[derive]	Implant-stem mismatch; potential intraoperative fracture or non-union - patient harm	8	[assess]	[assess]
Surface roughness Ra > 0.8 um on taper	[derive]	[derive]	Reduced osseointegration; long-term implant loosening - re-operation risk	9	[assess]	[assess]
Fixture slip / part shift mid-cycle	[derive]	[derive]	Geometric error cascade (runout, perpendicularity OOS); scrap or undetected escape	7	[assess]	[assess]
Coolant pressure drop / flow loss	[derive]	[derive]	Thermal damage to Ti substrate; metallurgical alteration; implant scrapped or escaped	9	[assess]	[assess]
Wrong program revision loaded (obsolete program)	[derive]	[derive]	Incorrect geometry machined; toolpath collision risk; patient dimension non-conformance	8	[assess]	[assess]
CMM inspection step skipped / bypassed	[derive]	[derive]	Non-conforming part released to sterilization and distribution - direct patient risk	7	[assess]	[assess]

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485 | ISO 14971:2019 - map P1xP2->Risk per device risk policy